STAT 641 **Midterm Exam 2 Solution** **Ghosh**

1. (a) (i) For all x, $f_n(x) = 0$ for large n. This proves $f_n \to f$ almost surely (m) as $n \to \infty$.
   (ii) Since for fixed $T > 1$, $\int |f_n| I_{\{|f_n|>T\}}\, \mathrm{d}m = 0,\ \forall\ n \geq 1$, $\{X_n\}$ is UI.
   (iii) Here $\int |f_n - f| \mathrm{d}m = \int |f_n| \mathrm{d}m = 1 \cdot m([n, n+1]) = 1$ for all n large, so f_n does not converge to f in L^1.

   (b) If $\{f_n\}$ is UI, and the measure is finite, then $f_n \to f$ a.e (μ) would imply $f_n \to f$ in L^1. There are other correct answers too, for example, along the lines of DCT.

2. (a) $\mu(\emptyset) = (0.1)\mu_1(\emptyset) + (0.7)\mu_2(\emptyset) + (0.2)\mu_3(\emptyset) = 0$; and $\mu(A) = (0.1)\mu_1(A) + (0.7)\mu_2(A) + (0.2)\mu_3(A) \geq 0 \Rightarrow \mu(A) \in [0, \infty],\ \forall\ A \in \mathcal{F}$; If $\{A_i\}$ is a disjoint collection of sets in $\mathcal{F}$, then $\mu(\cup_i A_i) = (0.1)\mu_1(\cup_i A_i) + (0.7)\mu_2(\cup_i A_i) + (0.2)\sum_i \mu_3(\cup_i A_i) = (0.1)\sum_i \mu_1(A_i) + (0.7)\sum_i \mu_2(A_i) + (0.2)\sum_i \mu_3(A_i) = \sum_i \mu(A_i)$. So μ is a measure. Also, $\mu(I\!R) = (0.1)\mu_1(I\!R) + (0.7)\mu_2(I\!R) + (0.2)\mu_3(I\!R) = 1$

   (b) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $[(0.7)\mu_2](B^c) = 0 = \nu(B)$, i.e. $(0.7)\mu_2 \perp \nu$. Choose $\mu_s = (0.7)\mu_2, \mu_a = (0.1)\mu_1 + (0.2)\mu_3$, then $\mu = \mu_a + \mu_s$. If $\nu(A) = 0$, then it follows that $\mu_a(A) = 0$. Hence, $\mu_a \ll \nu$.
   Computing: $\dfrac{d\mu_a}{d\nu}$: Note that if $\phi_{2,3}$ represent the density of $Normal(2,3)$, then by definition $\mu_1(A) = \int_A \phi_{2,3} dm$ and $\nu = m$. So, $\dfrac{d\mu_1}{d\nu} = \phi_{2,3}$ from RN theorem. Similarly, $\dfrac{d\mu_3}{d\nu} = f_{exp}$ = exponential (1) density function. Hence, from the proposition in the text, $\dfrac{d\mu_a}{d\nu} = \dfrac{d((0.1)\mu_1 + (0.2)\mu_3)}{d\nu} = (0.1)\dfrac{d\mu_1}{d\nu} + (0.2)\dfrac{d\mu_3}{d\nu} = (0.1)\phi_{2,3} + (0.2)f_{exp}$.

   (d) Using change of variable formula, we know:

$$E(X^2) = \int x^2 dP_X(x) = \int x^2 d\mu(x) = (0.1)\int x^2 d\mu_1(x) + (0.7)\int x^2 d\mu_2(x) + (0.2)\int x^2 d\mu_3(x)$$

   $= (0.1)(7) + (0.7)(42) + (0.2)(2) = 30.5$, since second moment of Normal(2, 3) is $(3+2^2) = 7$ and second moment of Poisson $(\lambda = 6)$ is $\lambda + \lambda^2 = 6 + 6^2 = 42$ and second moment of $Exponential(1)$ is $1 + 1^2 = 2$.

3. (a) Let $f(X) = |X|^{p_1}, g = 1, r = \frac{p_2}{p_1} > 1, q = \frac{r}{r-1}$ where $\frac{1}{r} + \frac{1}{q} = 1$.
   Since $E|f|^r = E|X|^{p_2} < \infty$ and $E|g|^q = E1 = 1 < \infty$, then by Hölder's inequality, $E|fg| \leq (E|f|^r)^{\frac{1}{r}}(E|g|^q)^{\frac{1}{q}}$, i.e. $E(|X|^{p_1} \cdot 1) \leq [E(|X|^{p_2})]^{\frac{1}{r}} \cdot 1$, i.e. $E(|X|^{p_1}) \leq [E(|X|^{p_2})]^{\frac{p_1}{p_2}}$. Since $E|X|^{p_2} < \infty$, then the above inequality gives $E|X|^{p_1} < \infty$, i.e. $X \in L^{p_1}(\Omega, \mathcal{F}, \mu)$.

   (b) If $X \in L^\infty$, then $||X||_\infty = \inf\{c > 0 : P(|X| > c) = 0\} < \infty$. So, there is $c_0 < \infty$ such that $P(|X| > c_0) = 0$ or $P(|X| \leq c_0) = 1$ or $|X| \leq c_0$ a.e. (P). So $E|X|^p < c_0^p < \infty$ for $0 < p < \infty$.